Proposed Methodology

1. Study Design and Data

1.1 Design

This retrospective, multi-cohort computational modeling study used the RASTA dataset. Model development followed a nested, patient-grouped evaluation design. The fixed outer folds provided held-out estimates; an inner patient-grouped split within each outer-training set supported early stopping, hyperparameter selection, and temperature calibration. Reporting followed applicable elements of the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis for Artificial Intelligence (TRIPOD+AI) and Checklist for Artificial Intelligence in Medical Imaging (CLAIM) guidance.

1.2 Cohorts and outcome encoding

The study includes the four original dataset folders with stable zero-based labels in folder-number order.

Label	Dataset folder	Analytical role
0	1_RETINORM	Source-cohort class
1	2_ORNET	Source-cohort class
2	3_FAMILIPO	Source-cohort class
3	4_MRCC	Source-cohort class; coronary revascularization-associated cohort

The primary outcome was folder-derived cohort identity. Because cohort and study membership coincide, performance was interpreted as cohort discrimination and not as proof of CAD detection, cardiovascular screening utility, or general clinical generalizability.

1.3 Unit of observation and notation

Let i index patients, e index eyes (OD or OS), and l index OCTA layers (SVC, DVC, or CC). An eye-level record contained three registered layer images, patient-level clinical variables, missingness indicators, and the cohort label. Eyes were used as training samples, but both eyes from a patient remained in the same partition. Final predictions, confidence intervals, and hypothesis tests used the patient as the independent unit.

1.4 Eligibility and quality control

Records required a resolvable patient identifier and at least the image inputs needed by the evaluated model. Images were screened for unreadable files, invalid dimensions, zero variance, all-black content, low dynamic range, and predefined quality failures. Exclusions and their reasons were recorded before modeling and summarized by cohort and eye. Missing modalities were not silently replaced. A model-specific complete-case rule or an explicit missing-modality indicator was declared for each experiment.

1.5 Clinical variables

Candidate clinical variables were drawn from the accompanying publication table and data manifest. Variable definitions, units, allowable ranges, coding, and missingness were documented before outcome modeling. Variables that deterministically defined the endpoint were excluded from predictors. Continuous variables were standardized; binary variables preserved their encoded state; ordinal variables used learned embeddings. All parameters and category vocabularies were fitted on the outer-training data only.

2. Computational Framework

Three OCTA layers → Fold-local contrastive pretraining → Layer-aware attention → Clinical, biomarker, and mask fusion → Calibrated patient-level output

Figure 1. High-level flow of the proposed framework. Every learned component and preprocessing estimate is fitted without access to the corresponding outer-test patients.

2.1 OCTA preprocessing

Each grayscale image was clipped using its second and ninety-eighth intensity percentiles, rescaled to [0,1], enhanced with contrast-limited adaptive histogram equalization, and resized to 224 × 224 pixels by bicubic interpolation. Fold- and layer-specific means and standard deviations were computed using only outer-training patients and applied to inner-validation and outer-test images.

$$x^{(p)} = \mathrm{clip}\left(\frac{x - q_2(x)}{q_{98}(x) - q_2(x) + \epsilon}, 0, 1\right), \qquad \hat{x}^{\ell} = \frac{x^{(p),\ell} - \mu_{\ell,f}}{\sigma_{\ell,f} + \epsilon}.$$

Geometric augmentation was synchronized across the three layers from the same eye to preserve spatial correspondence. Photometric augmentation varied by layer but remained conservative enough to preserve capillary morphology. Validation and test inputs were not augmented.

2.2 Missing clinical data

For clinical feature *j*, an observation indicator accompanied the standardized value. A learned feature-specific token replaced missing values inside the model. During training, 10% of observed entries were masked at random to improve robustness. Missingness rates were reported by cohort. Complete-case analysis and conventional training-fold median/mode imputation remain planned sensitivity ablations.

$$\tilde{c}_{ij} = r_{ij}\frac{c_{ij} - \mu_{j,f}}{\sigma_{j,f} + \epsilon} + (1 - r_{ij})a_j.$$

2.3 Layer-specific contrastive pretraining

Independent one-channel ConvNeXt-Tiny encoders were pretrained for SVC, DVC, and CC within each outer-training fold. For patients with two eyes, the same layer from OD and OS formed a natural positive pair. For unilateral records, two independently augmented views of the same image formed the pair. Direct comparison of the bilateral strategy with conventional same-image SimCLR remains a planned ablation because contralateral eyes may differ in pathology and artifact burden.

For augmented views v_i and v_j, layer-specific encoder $f_{\theta l}$, projection head $g_{\varphi l}$, and L2-normalized projection z, the representation and pairwise similarity are defined as follows:

$$z_i = \frac{g_{\phi_\ell}(f_{\theta_\ell}(v_i))}{\|g_{\phi_\ell}(f_{\theta_\ell}(v_i))\|_2}, \qquad s_{ik} = z_i^\top z_k.$$

For a minibatch containing B positive pairs and 2B views, the directional normalized temperature-scaled cross-entropy loss and its symmetric batch average are

$$\ell_{i,j} = -\log \frac{\exp(s_{ij}/\tau)}{\sum_{k=1}^{2B} \mathbb{1}_{[k\neq i]} \exp(s_{ik}/\tau)}, \qquad \mathcal{L}_{\mathrm{NTX}} = \frac{1}{2B} \sum_{(i,j)} \ell_{i,j}.$$

Each encoder was followed by a two-layer projection head and optimized with the normalized temperature-scaled cross-entropy objective. The four-class run used 100 epochs, temperature 0.07, SGD with momentum 0.9, weight decay 10^{-4}, initial learning rate 0.03 with cosine decay, effective batch size 256, and gradient clipping at 1.0. Preflight diagnostics checked finite gradients, parameter updates, embedding variance, and separation from the random-loss baseline. Projection heads were discarded after pretraining.

2.4 Layer-specific feature extraction

Each pretrained backbone produces a 7 × 7 × 768 feature map. Global average pooling followed by a layer-specific projection and layer normalization yields a 256-dimensional token. Final convolutional maps are retained for Grad-CAM. The three backbones remain independent so that depth-specific appearance is not forced into a single shared representation.

$$u^\ell = \mathrm{GAP}(U^\ell), \qquad f^\ell = \mathrm{LN}\big(W_2^\ell\, \mathrm{GELU}(W_1^\ell u^\ell + b_1^\ell) + b_2^\ell\big) \in \mathbb{R}^{256}.$$

2.5 Identity-aware cross-layer attention

Learned identity embeddings are added to the SVC, DVC, and CC tokens before an eight-head self-attention block. Each head has dimension 32; the feed-forward width is 1024 and dropout is 0.1. Residual connections and pre-normalization are used. The three output tokens are mean-pooled to obtain the 256-dimensional OCTA representation.

$$T = \begin{bmatrix} f^S + e^S \\ f^D + e^D \\ f^C + e^C \end{bmatrix}, \qquad A_h = \mathrm{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{d_h}}\right), \qquad f_{\mathrm{octa}} = \frac{1}{3} \sum_{\ell \in \{S,D,C\}} H_\ell.$$

$$Q_h = \mathrm{LN}(T) W_h^Q, \quad K_h = \mathrm{LN}(T) W_h^K, \quad V_h = \mathrm{LN}(T) W_h^V, \quad O = \mathrm{Concat}_{h=1}^{8}(A_h V_h) W^O.$$

$$T' = T + \mathrm{Dropout}(O), \qquad H = T' + \mathrm{Dropout}\big(W_2\, \mathrm{GELU}(W_1\, \mathrm{LN}(T'))\big).$$

2.6 Generated masks and vascular biomarkers

Classical image-processing procedures produced SVC and DVC vessel masks, SVC and DVC foveal avascular zone (FAZ) masks, and a CC flow-deficit mask. These masks were pseudo-labels rather than human annotations. Quantitative thresholds and visual montages were used for quality control. A U-Net++ student was trained against QC-passed pseudo-labels within the permitted training data and then frozen before downstream classification. The operational downstream inputs were the frozen student outputs.

For pixel probability p_q, binary pseudo-label y_q, and pixel domain Ω, the segmentation losses are

$$\mathrm{Dice} = \frac{2\sum_{q\in\Omega} p_q y_q + \epsilon}{\sum_{q\in\Omega} p_q + \sum_{q\in\Omega} y_q + \epsilon}, \qquad \mathcal{L}_{\mathrm{Dice}} = 1 - \mathrm{Dice}.$$

$$\mathrm{TI} = \frac{TP}{TP + \alpha_{tv}FP + \beta_{tv}FN + \epsilon}, \qquad \mathcal{L}_{\mathrm{Tversky}} = 1 - \mathrm{TI}, \qquad \beta_{tv} = 0.7.$$

$$\mathcal{L}_{\mathrm{FocalSeg}} = -\frac{1}{|\Omega|}\sum_{q\in\Omega}\left[y_q(1-p_q)^{\gamma}\log p_q + (1-y_q)p_q^{\gamma}\log(1-p_q)\right], \qquad \gamma = 2.$$

$$\mathcal{L}_{\mathrm{student}} = \alpha_s\mathcal{L}_{\mathrm{Tversky}} + (1-\alpha_s)\mathcal{L}_{\mathrm{FocalSeg}} + \lambda_d\mathcal{L}_{\mathrm{Dice}}.$$

Derived biomarkers included vessel density, vessel length density, FAZ area, FAZ circularity, normalized FAZ displacement, CC flow-deficit percentage, connected-component count, and mean and median deficit area. Physical areas used the available pixel-scale metadata. Implausible or failed measurements were flagged rather than coerced to zero. The five masks were stacked as a five-channel 224 × 224 input and encoded into a 128-dimensional mask representation.

$$\mathrm{VD} = \frac{\sum_{q\in\Omega} M_q}{|\Omega|}, \quad \mathrm{VLD} = \frac{\sum_{q\in\Omega} S(M)_q}{|\Omega|}, \quad A_{FAZ} = a_{px}|F|, \quad C_{FAZ} = \frac{4\pi A_{FAZ}}{P_{FAZ}^2 + \epsilon}.$$

$$d_{FAZ} = \frac{\|\bar{p}_F - p_{center}\|_2}{\sqrt{H^2 + W^2}}, \qquad \mathrm{FD}\,\% = 100\frac{\sum_{q\in\Omega} D_q}{|\Omega|}.$$

$$M_{all} = \mathrm{Stack}(M_S, M_D, F_S, F_D, D_C) \in \mathbb{R}^{5\times224\times224}, \qquad f_{mask} = E_m(M_{all}) \in \mathbb{R}^{128}.$$

2.7 Tabular encoder and multimodal fusion

The clinical vector and normalized generated biomarkers were concatenated and processed by a two-layer multilayer perceptron with GELU activations, layer normalization, and dropout, producing a 256-dimensional tabular representation. The tabular vector conditioned the OCTA representation through a learned projection and residual addition. This conditioned vector was then concatenated with the 128-dimensional mask representation and projected to a 512-dimensional fused representation. Each modality was included once to preserve interpretability of the planned ablations.

$$q = [\tilde{c}; b], \quad h_1 = \mathrm{Drop}_{0.3}(\mathrm{GELU}(W_1\,\mathrm{LN}(q) + b_1)), \quad f_{tab} = \mathrm{Drop}_{0.2}(\mathrm{LN}(\mathrm{GELU}(W_2 h_1 + b_2))) \in \mathbb{R}^{256}.$$

$$f_{cond} = \mathrm{LN}(f_{octa} + W_t f_{tab} + b_t), \qquad h = \mathrm{LN}(\mathrm{GELU}(W_f[f_{cond}; f_{mask}] + b_f)) \in \mathbb{R}^{512}.$$

2.8 Prototype-augmented classifier

A two-layer classification head produced a 128-dimensional penultimate embedding and four parametric logits. During training, an exponential-moving-average prototype was maintained for every class from the embeddings observed for that class. Negative squared Euclidean distances to prototypes formed auxiliary logits, scaled by a positive learned coefficient and combined with the parametric logits. Prototype state was estimated from training data only and reset independently for every outer fold.

$$z = \mathrm{Drop}_{0.3}(\mathrm{GELU}(W_2\,\mathrm{Drop}_{0.3}(\mathrm{GELU}(W_1 h + b_1)) + b_2)), \qquad s = W_3 z + b_3.$$

$$\bar{z}_k^{(t)} = \frac{1}{|B_k|} \sum_{i \in B_k} z_i, \qquad P_k^{(t)} = \rho P_k^{(t-1)} + (1-\rho)\bar{z}_k^{(t)}.$$

$$r_k = -\frac{\|z - P_k\|_2^2}{\tau_p}, \quad \tau_p = 0.1, \qquad o_k = s_k + \mathrm{softplus}(a) r_k, \qquad p_k = \frac{\exp(o_k)}{\sum_{j=1}^{4} \exp(o_j)}.$$

2.9 Classification objective and imbalance

The default objective was inverse-frequency-weighted cross-entropy with label smoothing $\varepsilon = 0.1$. Class weights were computed from outer-training patients and normalized to mean one. Focal cross-entropy with $\gamma = 2$ remains a planned ablation for training folds with an imbalance ratio exceeding 10:1. Patient-balanced sampling was used so that patients with two eyes did not receive twice the sampling weight of unilateral patients.

$$\tilde{y}_{ik} = (1-\varepsilon)\mathbb{1}[y_i = k] + \frac{\varepsilon}{K}, \quad K = 4, \qquad w_k = \frac{1/n_k}{\sum_j 1/n_j} K.$$

$$\mathcal{L}_{cls} = -\frac{1}{B} \sum_{i=1}^{B} \sum_{k=1}^{K} w_k \tilde{y}_{ik} \log(p_{ik} + \epsilon).$$

$$\mathcal{L}_{focal} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{k=1}^{K}\tilde{y}_{ik}(1-p_{ik})^{\gamma}\log(p_{ik}+\epsilon), \qquad \gamma = 2.$$

2.10 Training schedule

Stage	Trainable components	Optimization
1. Representation adaptation	Three independent layer encoders and projection heads	Fold-local SimCLR, 100 epochs; SGD; cosine schedule
2. Fusion warm-up	Projection, attention, tabular, mask, fusion, prototype, and classifier modules; image backbones frozen	15 epochs; learning rate 3×10^{-4}
3. End-to-end fine-tuning	All classifier components; segmentation student remains frozen	AdamW; backbone 3×10^{-5}; downstream 3×10^{-4}; weight decay 10^{-2}; 5 warm-up epochs; cosine annealing; maximum 100 epochs

Early stopping used inner-validation macro-F1 with patience 15. Gradient norms, learning rates, training losses, validation metrics, and random seeds were recorded. Checkpoints were selected without reference to outer-test performance.

3. Evaluation and Statistical Analysis

3.1 Nested patient-grouped evaluation

Five fixed patient-grouped folds served as outer test partitions. Within each outer-training set, an 80/20 patient-grouped split, stratified when feasible, was used for model selection and calibration. No outer-test patient contributed to image statistics, clinical preprocessing, contrastive pretraining, pseudo-label model selection, feature selection, early stopping, prototype construction, or temperature fitting.

Eye-level probabilities were averaged within patient before primary evaluation. Comparison with logit averaging and quality-weighted averaging remains a planned aggregation ablation.

$$\bar{p}_{ik} = \frac{1}{|E_i|}\sum_{e \in E_i} p_{iek}, \qquad \hat{y}_i = \arg\max_k \bar{p}_{ik}.$$

3.2 Outcomes and metrics

The primary metric was patient-level macro-F1 pooled from out-of-fold predictions. Secondary discrimination metrics were balanced accuracy, weighted F1, macro one-vs-rest ROC-AUC, macro PR-AUC, top-2 accuracy, and per-class sensitivity, specificity, precision, F1, and support. Normalized and count confusion matrices were reported. Macro metrics were emphasized because class frequencies were unequal and performance in smaller cohorts needed to remain visible.

$$\text{Sensitivity}_k = \frac{TP_k}{TP_k + FN_k}, \quad \text{Specificity}_k = \frac{TN_k}{TN_k + FP_k}, \quad \text{Precision}_k = \frac{TP_k}{TP_k + FP_k}.$$

$$F1_k = \frac{2\,\mathrm{Precision}_k\,\mathrm{Sensitivity}_k}{\mathrm{Precision}_k + \mathrm{Sensitivity}_k}, \qquad \mathrm{MacroF1} = \frac{1}{K}\sum_{k=1}^{K} F1_k.$$

$$\mathrm{BalancedAccuracy} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{Sensitivity}_k, \qquad \mathrm{WeightedF1} = \sum_{k=1}^{K} \frac{n_k}{N} F1_k.$$

Calibration was summarized using multiclass negative log-likelihood, Brier score, expected calibration error with prespecified bins, classwise calibration plots, and reliability diagrams. Metrics were reported before and after calibration.

$$\mathrm{Brier} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}(\bar{p}_{ik} - \mathbb{1}[y_i = k])^2, \qquad \mathrm{ECE} = \sum_{g=1}^{G} \frac{|B_g|}{N} |\,\mathrm{acc}(B_g) - \mathrm{conf}(B_g)|.$$

3.3 Calibration and uncertainty

A single positive temperature was fitted to inner-validation logits by minimizing negative log-likelihood and then applied unchanged to the associated outer-test logits. Epistemic uncertainty was approximated using 30 Monte Carlo dropout passes. Predictive entropy and class-probability variance were recorded. The uncertainty threshold for referral was selected on inner-validation data. Outer-test reporting included accuracy by entropy stratum, risk-coverage curves, coverage at prespecified risk levels, and the association between uncertainty and error.

$$T^* = \arg\min_{T>0} \left[-\sum_i \log \mathrm{softmax}(o_i/T)_{y_i} \right].$$

$$\bar{p}_k = \frac{1}{M}\sum_{m=1}^{M} p_k^{(m)}, \qquad \widehat{\mathrm{Var}}(p_k) = \frac{1}{M-1}\sum_{m=1}^{M}(p_k^{(m)} - \bar{p}_k)^2, \qquad M = 30.$$

$$H(\bar{p}) = -\sum_{k=1}^{K} \bar{p}_k \log(\bar{p}_k + \epsilon), \qquad \mathrm{Coverage}(\eta) = \frac{1}{N}\sum_i \mathbb{1}[H(\bar{p}_i) \le \eta].$$

3.4 Confidence intervals and comparisons

Ninety-five percent confidence intervals were estimated from 1,000 stratified patient-level bootstrap resamples of the pooled out-of-fold predictions. Model comparisons used paired resampling on identical patients. DeLong tests were used for paired AUC comparisons where assumptions were satisfied, and McNemar tests compared paired classification errors. Confirmatory comparisons were prespecified and adjusted for multiplicity using Holm correction. Effect sizes and confidence intervals accompanied p-values.

$$CI_{95\%} = \left[Q_{0.025}(\{\hat{\theta}^{*(b)}\}), Q_{0.975}(\{\hat{\theta}^{*(b)}\})\right], \qquad b = 1, \ldots, 1000.$$

$$\chi^2_{\mathrm{McNemar}} = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}}.$$

3.5 Planned baselines and ablations

Question	Planned comparison
Value of multimodal learning	Clinical-only logistic regression, random forest, and XGBoost; image-only model; image plus clinical; image plus masks; complete model
Value of retinal depth	SVC-only, DVC-only, CC-only, three layers stacked as channels, independent encoders with direct concatenation, and identity-aware attention
Value of pretraining	No contrastive pretraining, standard same-image SimCLR, and bilateral-positive SimCLR
Value of generated structure	No biomarkers; no masks; classical pseudo-masks; frozen U-Net++ masks
Value of class prototypes	Parametric classifier alone versus prototype-augmented classifier
Value of uncertainty processing	Uncalibrated versus temperature-scaled; deterministic versus MC-dropout inference
Published-style image baseline	EfficientNetV2-B3 trained and evaluated under the same patient-grouped partitions

3.6 Confounding, leakage, and robustness analyses

- Clinical-only cohort classification was measured. Strong performance was treated as evidence of selection effects requiring cautious interpretation.
- Image-only analyses were repeated after age and sex matching or inverse-probability weighting where overlap and sample size permitted.
- The predictiveness of image-quality measures, missingness patterns, and available acquisition metadata was assessed.
- Unilateral patients, bilateral patients, and individual eyes were compared; agreement between eyes was calculated, and patient-clustered inference was used throughout.
- Performance was evaluated by sex, age group, diabetes status, eye laterality, image-quality stratum, and cohort size, subject to minimum subgroup counts.
- Analyses were repeated after excluding failed or borderline pseudo-mask cases and after replacing learned missing tokens with conventional imputation.
- Fold-level heterogeneity was inspected so that a pooled score did not conceal a severe failure in one outer fold or one cohort.

3.7 Explainability analysis

Grad-CAM was generated from the final convolutional feature maps for each layer. Cross-layer attention matrices were summarized as incoming attention by retinal depth. SHAP was applied to the tabular branch using a training-derived background set. Explanations were reviewed for anatomical plausibility, border artifacts, acquisition markings, and cohort-specific shortcuts. These analyses were descriptive and were not interpreted as causal evidence. Interpretation was accepted only when consistent with error analysis and will be revisited after the planned matched ablations.

$$\alpha_r^c = \frac{1}{Z}\sum_{u,v}\frac{\partial o_c}{\partial A_{uv}^r}, \qquad L_{\mathrm{GradCAM}}^c = \mathrm{ReLU}\left(\sum_r \alpha_r^c A^r\right).$$

$$d_k = \frac{1}{3}\sum_{q=1}^{3} A_{qk}, \qquad k \in \{\mathrm{SVC}, \mathrm{DVC}, \mathrm{CC}\}.$$

4. Reproducibility and Data Governance

Every experiment stored the software environment, configuration, seed, fold identifiers, preprocessing parameters, class weights, selected epoch, calibration temperature, and checkpoint hash. Data manifests and derived outputs preserved patient identifiers only in controlled, non-public form; reports used aggregate counts and de-identified examples.

Intermediate and final artifacts were organized by phase and outer fold. Each model result was reconstructable from a frozen configuration and immutable input manifest. Failed runs were retained in the audit trail. Code used for final analysis was versioned before test-fold aggregation, and all reported metrics were regenerated from saved out-of-fold patient predictions.